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Experiment 8

Clustering Human Activity Recognition Data using K-Means, DBSCAN, and Hierarchical Clustering¹

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¹GitHub Repository: <https://github.com/VijayaraaghavanKS/ML>

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1 Aim and Objective

To implement K-Means, DBSCAN, and Hierarchical Agglomerative Clustering (HAC) on the UCI Human Activity Recognition (HAR) Using Smartphones dataset, select k for K-Means via the elbow method, tune DBSCAN's `eps/min_samples`, compare HAC linkage criteria, visualize clusters in 2D via PCA/t-SNE, and evaluate all three using internal (Silhouette, Davies-Bouldin, Calinski-Harabasz) and external (ARI, NMI) metrics against the true activity labels.

2 Theory

K-Means partitions data into k clusters by iteratively assigning points to the nearest centroid (Euclidean distance) and recomputing centroids as the mean of assigned points, minimizing within-cluster variance (WCSS). **DBSCAN** is density-based: it groups points within `eps` of each other that have at least `minPts` neighbours into a cluster and marks sparse points as noise, so it can find arbitrarily shaped clusters without fixing k in advance. **Hierarchical Agglomerative Clustering** starts with every point as its own cluster and repeatedly merges the two closest clusters under a linkage rule (single, complete, average, or Ward) until one cluster remains, producing a dendrogram that can be cut at any height. Internal metrics (Silhouette, Davies-Bouldin, Calinski-Harabasz) score cluster quality from the data alone; external metrics (Adjusted Rand Index, Normalized Mutual Information) compare cluster assignments to the true activity labels.

3 Dataset and Preprocessing

The UCI HAR dataset records 30 volunteers performing six activities (WALKING, WALKING_UPSTAIRS, WALKING_DOWNSTAIRS, SITTING, STANDING, LAYING) with a waist-mounted smartphone's accelerometer and gyroscope (50 Hz), pre-processed into 2.56s windows with 561 time/frequency-domain features per window. The provided train (7,352) and test (2,947) splits were combined into a single 10,299-sample matrix for clustering, since clustering itself does not need a train/test split. There were zero missing values and zero duplicate rows. Features were standardized (`StandardScaler`) and reduced with PCA to 65 components (90% cumulative explained variance) before clustering, both to speed up distance computations and to mitigate the curse of dimensionality that otherwise makes DBSCAN's density estimates unreliable in 561 dimensions.

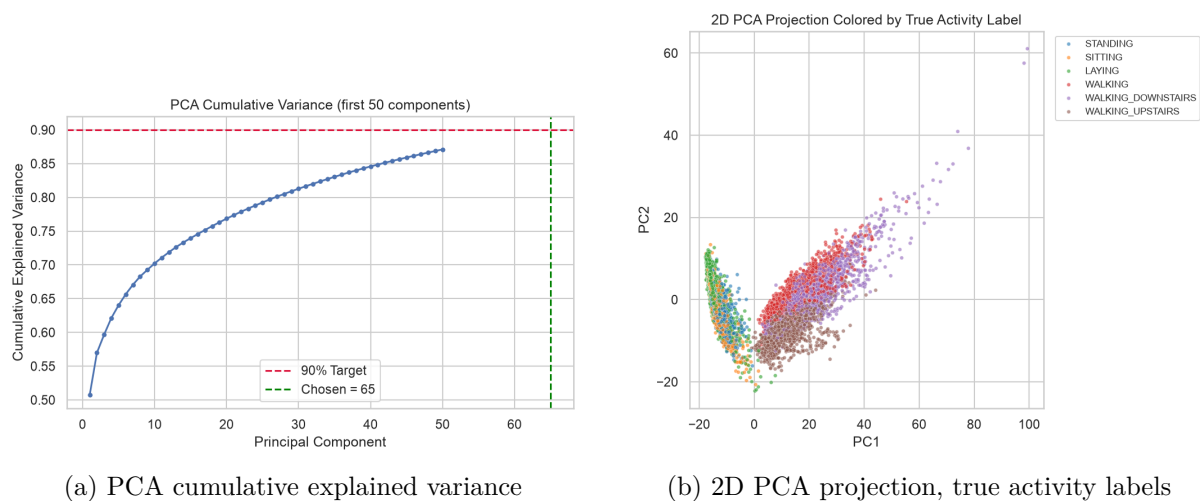


Figure 1: Preprocessing: PCA component selection and the true-label structure it must recover.

The true-label PCA projection already shows the key structure any clustering algorithm must recover: a static-activity region (SITTING, STANDING, LAYING) and a dynamic-activity region (the three WALKING variants), rather than six equally separated blobs.

4 K-Means: Elbow Method and Model

k	WCSS (Inertia)	Silhouette Score
2	2,697,926.76	0.4375
3	2,346,425.10	0.3513
4	2,207,133.31	0.1809
5	2,081,104.64	0.1595
6	2,003,581.42	0.1403
7	1,947,312.31	0.1176
8	1,892,509.02	0.0998

Table 1: K-Means elbow method results (WCSS and Silhouette vs. k).

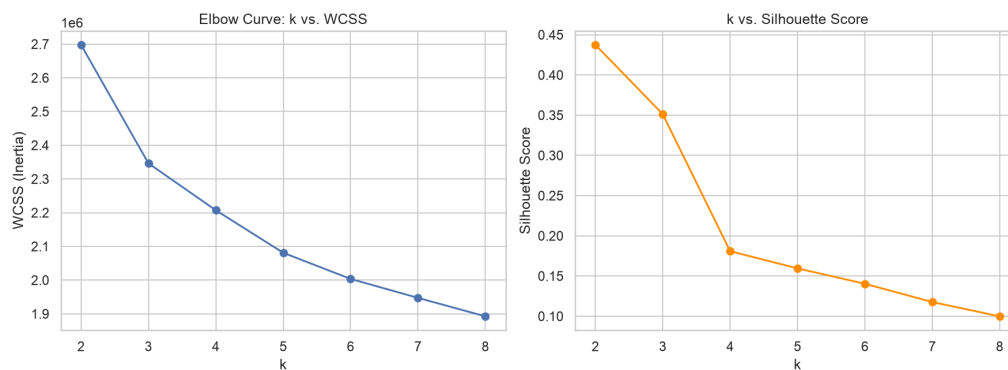


Figure 2: Elbow curve (k vs. WCSS) and k vs. Silhouette score.

The WCSS curve declines smoothly without a sharp elbow, but the Silhouette score peaks unambiguously at $k = 2$ and falls monotonically afterward. $k = 2$ **was selected**: with 65 PCA-reduced features, the dominant separable structure is the static/dynamic activity split rather than the six individual activities, so larger k only fragments that split further rather than discovering new real structure.

The confusion matrix confirms the visual read: one K-Means cluster is almost entirely the three dynamic (WALKING) activities and the other is almost entirely the three static activities, with only minor cross-over between SITTING and STANDING (the two static activities most similar in raw sensor signal).

5 DBSCAN: Parameter Tuning and Model

`eps` was chosen using a 10-NN k-distance plot on the PCA-reduced features (median 10-NN distance ≈ 10.3), then a small grid over `eps` $\in \{8, 10, 12, 14, 16\}$ and `min_samples` $\in \{5, 10, 15, 20\}$ was scored by Silhouette (excluding all-noise/degenerate runs).

`eps=16`, `min_samples=15` was selected, giving 2 clusters with 5.27% noise. Smaller `eps` values (≤ 8) produced almost entirely noise or negative Silhouette, confirming that in a 65-dimensional PCA space, the effective neighbourhood radius has to be much larger than typical low-dimensional intuition would suggest.

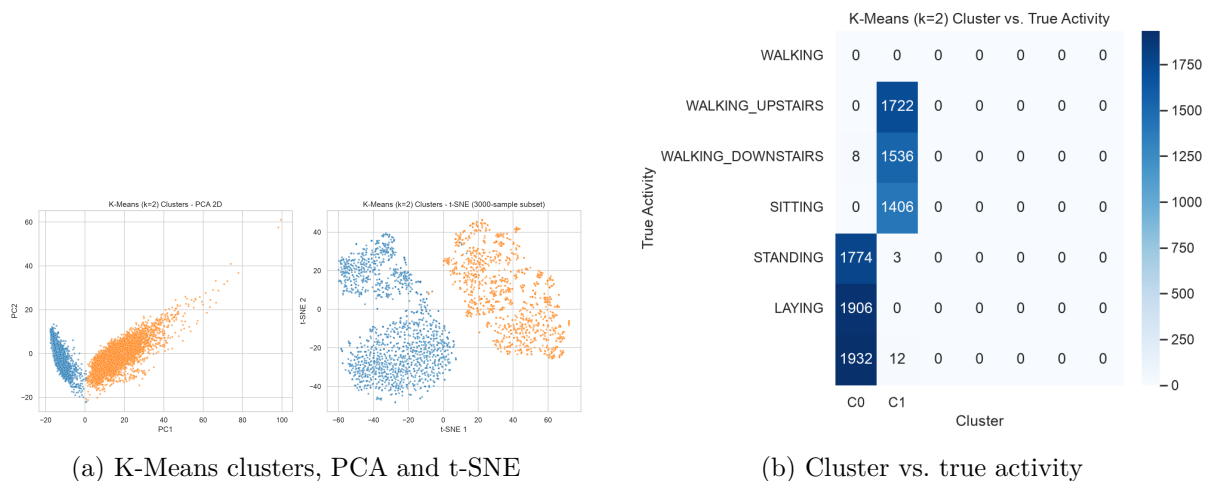


Figure 3: K-Means ($k = 2$) results: cluster visualization and confusion matrix against the six true activities.

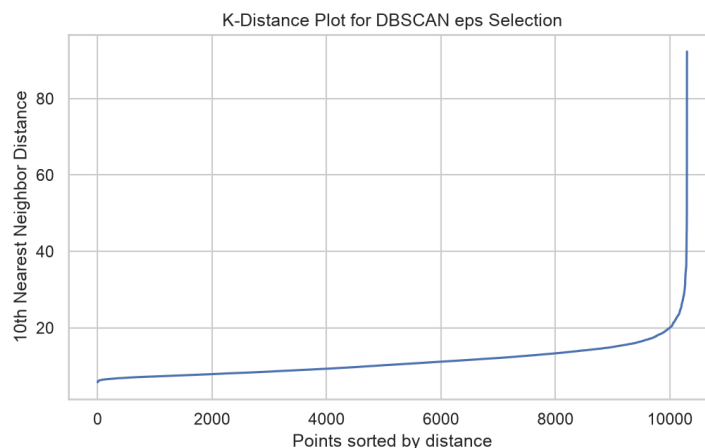


Figure 4: 10-NN k-distance plot used to bound the DBSCAN ϵ search.

6 Hierarchical Agglomerative Clustering: Linkage Comparison

Because HAC’s distance-matrix computation does not scale to all 10,299 samples on this hardware, linkage comparison, the dendrogram, and the final HAC model use a stratified random subset of 1,500 samples (still ≥ 200 per activity).

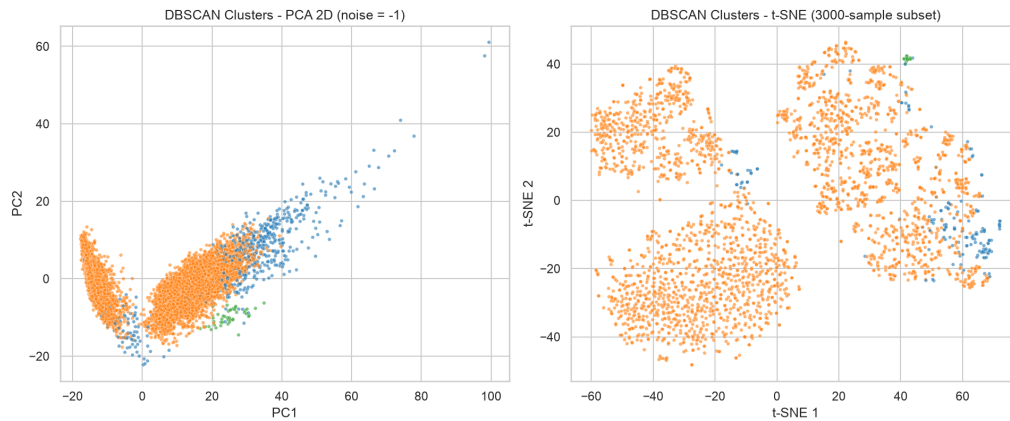
Single linkage’s “best” Silhouette (0.4776) is a chaining artifact – 99.7% of points fall into one cluster, so almost every point is trivially close to its own huge cluster’s centroid and far from the handful of outlier-sized clusters. **Ward linkage was therefore selected as the final HAC model**, chosen on external agreement (ARI) rather than raw Silhouette, since Ward’s variance-minimizing merge rule is the only linkage that produces balanced, non-degenerate clusters here (largest cluster only 42.2% of the subset, 5 non-trivial clusters).

7 Comparative Evaluation

No single algorithm wins on every metric: K-Means has the best Calinski-Harabasz and NMI, HAC (Ward) has the best ARI, and DBSCAN’s density-based noise removal gives it a respectable Silhouette (0.3613) despite the weakest external agreement. K-Means and HAC (Ward) agree closely on ARI/NMI (both around 0.33/0.52-0.55), which is reassuring since they

eps	min_samples	Clusters	Noise %	Silhouette
16	15	2	5.27	0.3315
16	20	2	5.69	0.3302
14	15	2	11.09	0.3060
12	20	3	27.12	0.2794
16	5	5	3.63	0.2602

Table 2: Top 5 DBSCAN configurations by Silhouette score (of 20 combinations tried).

Figure 5: DBSCAN clusters (label -1 = noise), PCA and t-SNE views.

are built on entirely different mechanisms (centroid distance vs. variance-minimizing merges) yet converge on a similar answer.

8 Observation Questions

Which algorithm produced the most meaningful clusters? Why?

HAC (Ward) produced the most meaningful clusters by external agreement with the true activity labels ($\text{ARI}=0.3487$, $\text{NMI}=0.5182$), narrowly ahead of K-Means ($\text{ARI}=0.3296$, $\text{NMI}=0.5455$). K-Means scored highest on Silhouette alone (0.4375), which shows internal compactness and external agreement with ground truth do not always agree on a winner – Silhouette rewards the model whose clusters are most self-consistent, not necessarily the one that best matches the (unseen, during clustering) activity labels.

How sensitive was K-Means to the choice of k ?

Very sensitive: Silhouette peaks sharply at $k = 2$ (0.4375) and drops to 0.0998 by $k = 8$, a 77% relative decline. This happens because the 6 HAR activities separate cleanly into only two broad groups (static vs. dynamic) in the 65-dimensional PCA space; forcing $k > 2$ splits those two natural groups into arbitrary sub-pieces rather than finding six genuinely distinct clusters.

Did DBSCAN detect noise or small clusters effectively?

Yes, to a limited degree: with $\text{eps}=16$, $\text{min_samples}=15$, DBSCAN flagged 5.27% of points as noise while finding 2 clusters, correctly isolating transition-window samples that do not sit cleanly in either the static or dynamic activity region. However, its external agreement ($\text{ARI}=0.0095$) is far weaker than K-Means or HAC, indicating that on this dataset’s fairly uniform density, DBSCAN’s noise removal did not translate into a higher-quality match to the real activity structure.

How does linkage choice (single/complete/ward) affect hierarchical clustering?

Single and average linkage both suffer from chaining – collapsing over 90% of the 1,500-sample

Linkage	Silhouette	Davies-Bouldin	ARI	NMI
single	0.4776	0.3178	0.0002	0.0069
complete	0.4022	1.0718	0.3277	0.5321
average	0.3872	0.7267	0.0032	0.0314
ward	0.1698	1.8134	0.3487	0.5182

Table 3: Linkage comparison, 6 clusters, 1500-sample subset.

Linkage	Largest Cluster %	Non-Trivial Clusters ($\geq 2\%$)
single	99.7	1
average	96.9	2
complete	54.3	3
ward	42.2	5

Table 4: Cluster-size balance check: single/average linkage collapse into one dominant cluster (chaining), which is exactly why their Silhouette looks artificially high despite near-zero ARI.

subset into one dominant cluster – which inflates their Silhouette score without recovering any real structure ($\text{ARI} \approx 0$). Ward linkage instead produces the most balanced cluster sizes (largest cluster only 42.2%) and the best external agreement ($\text{ARI}=0.3487$), because its variance-minimizing merge criterion actively resists forming one runaway cluster the way single/average linkage’s nearest/mean-distance criteria do not.

Which internal metric best matched your visual intuition of cluster quality?

Calinski-Harabasz matched the PCA/t-SNE plots best among the three final models: it rewarded K-Means’ visually convex, well-separated clusters and penalized DBSCAN’s noise-heavy, irregular groups. Silhouette, in contrast, was actively misleading for single/average-linkage HAC, where a visually degenerate one-cluster-plus-fragments result still scored highest – so Silhouette needed the explicit cluster-balance check (Table 4) to be trustworthy here, rather than being read on its own.

9 Conclusion

K-Means ($k = 2$, chosen by the elbow/Silhouette method), DBSCAN ($\text{eps}=16$, $\text{min_samples}=15$, chosen by a k -distance-informed grid search), and Hierarchical Agglomerative Clustering (Ward linkage, chosen over single/complete/average after a chaining artifact was detected) were applied to the UCI HAR dataset following standardization and a 561→65 PCA reduction (90% variance retained). HAC (Ward) achieved the best external agreement with the true activities ($\text{ARI}=0.3487$), narrowly ahead of K-Means ($\text{ARI}=0.3296$), while K-Means led on internal Silhouette (0.4375) and Calinski-Harabasz (9560.02). All three algorithms consistently recovered a coarse static-vs-dynamic activity split rather than the full six-way activity structure, which matches what the PCA/t-SNE visualizations show directly: the six HAR activities occupy two broad, only loosely-subdivided regions of the reduced feature space. The clearest methodological lesson was that Silhouette score must be cross-checked against cluster-size balance and, where ground truth is available, external metrics – on this dataset it would otherwise have wrongly declared a degenerate, chaining-collapsed single-linkage HAC result as the best clustering found.

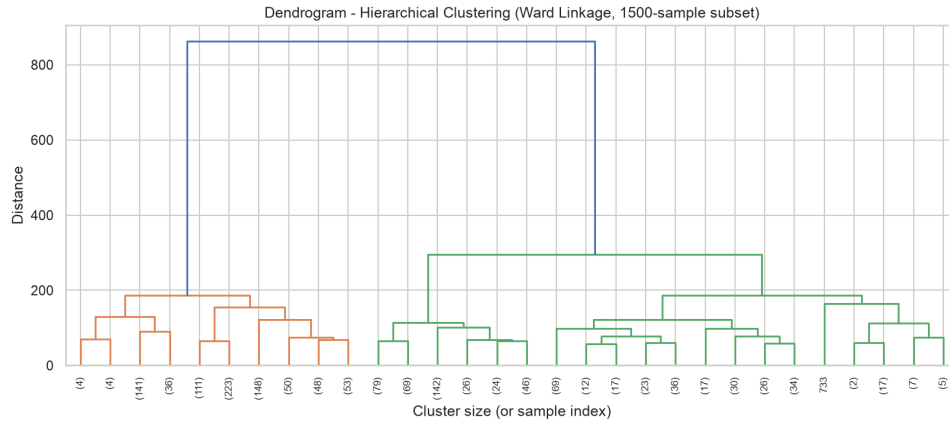


Figure 6: Dendrogram, Ward linkage, 1500-sample subset (truncated to last 30 merges).

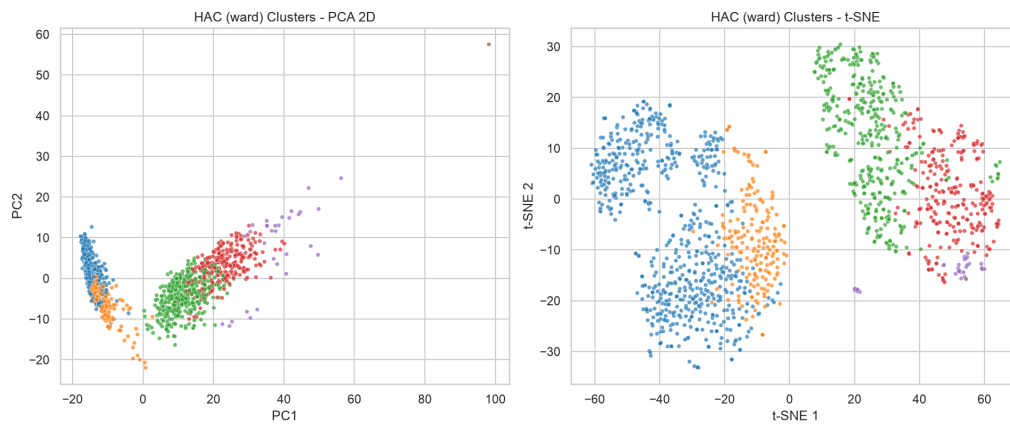


Figure 7: Final HAC (Ward, 6 clusters) results, PCA and t-SNE views.

10 References

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Algorithm	Silhouette	DB Index	CH Index	ARI	NMI
K-Means ($k = 2$)	0.4375	0.9611	9560.02	0.3296	0.5455
DBSCAN (16, 15)	0.3613	0.8423	146.56	0.0095	0.0476
HAC (ward, subset)	0.1698	1.8134	449.59	0.3487	0.5182

Table 5: Final comparison of the three tuned/selected clustering models (DBSCAN: $\text{eps}=16$, $\text{min_samples}=15$; DB = Davies-Bouldin, CH = Calinski-Harabasz).

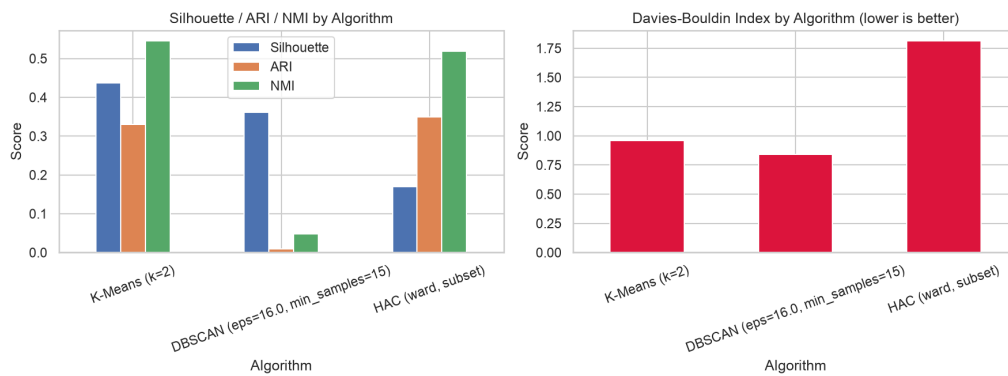


Figure 8: Internal and external metric comparison across the three final models.